Data Analytics

ECMP 5005D (Winter 2026)

Course Instructor: Esam Mahdi

Email: EsamMahdi@cunet.carleton.ca

Class Times and Location: Wednesday, 2:35 pm - 5:25 pm at Southam Hall (SA) room 409

Office Hours: TBD

Important Dates:

First day of classes: Jan. 5, 2026

Last day of classes: Apr. 8, 2026 (Classes follow a Friday schedule.)

Feb. 16–20, 2026: Winter break, no classes

Fri., Apr. 3, 2026: Statutory holiday, University closed

Course Outline Prerequisites:

Enrollment in the Master of Engineering Practice program

Course objectives

1. Develop skills to explore big data sets and identify patterns using data visualization techniques.
2. Develop the ability to interpret analytical results and communicate findings effectively to both technical and non-technical audiences through data storytelling and data visualization.
3. Compare and contrast supervised learning and unsupervised learning.
4. Use supervised and unsupervised machine learning algorithms to learn more about the data and build predictive models.
5. Apply data science to real-world problems in various fields and explore the challenges of working with massive data.

6. Acquire proficiency in R programming language and use relevant packages for data gathering, cleaning, manipulating, and analysing.

Course Description

Introduction to data analytics, including visualization and knowledge discovery in massive datasets; unsupervised learning: clustering algorithms; dimension reduction; supervised learning: pattern recognition, smoothing techniques, classification. Computer software will be used.

Texts & Course Materials:

All course materials (e.g., lecture slides/announcement/suggested questions etc.) will be available on Brightspace and links of this course. For security reasons, please do not copy or share contents anywhere. The majority of course communication and announcements will happen through Brightspace. It is your responsibility to check Brightspace regularly.

Students are not required to purchase any textbooks or learning materials for this course. The required textbook, An Introduction to Statistical Learning: With Applications in R (2021), is provided for free by the authors. This book is available for free on the instructors' website at https://www.statlearning.com/. The authors of the book are Gareth James, Daniela Witten, Trevor Hastie, and Robert Tibshirani, and it is published by Springer. Additionally, students can access a related online resource at https://r4ds.github.io/bookclub-islr/index.html.

Other References:

- *Robert I. Kabacoff. R in Action: Data analysis and graphics with R and Tidyverse. 3rd ed., Manning, 2022.*
- *Rafael A. Irizarry. Data Analysis and Prediction Algorithms with R. 1st ed., Chapman & Hall/CRC Data Science Series, 2020. https://rafalab.dfci.harvard.edu/dsbook/*
- *The Johns Hopkins Data Science Lab (DaSL): https://jhudatascience.org/index.html.*
- *Hadley Wickham, Garrett Grolemund. R for Data Science: Import, Tidy, Transform, Visualize, and Model Data. O'Reilly Media, Inc., 2016.*
- *H. Wickham. ggplot2 - Elegant Graphics for Data Analysis. Published in Use R! 3 October 2009. https://ggplot2-book.org/*
- *Claus Wilke. Fundamentals of Data Visualization: A Primer on Making Informative and Compelling Figures. 1st Ed. O'Reilly Media Inc., 2019. https://clauswilke.com/dataviz/*

Tentative Course Outline

Topic 1: Introduction to R, Rstudio, and RMarkdown.

Topic 2: Data visualization and interactive plots.

Topic 3: Linear regression (simple and multiple linear models).

Topic 4: Resampling methods (training/test sets, cross validation, bootstrap).

Topic 5: Classification (logistic regression, linear discriminant analysis, quadratic discriminant analysis, naive Bayes, k-nearest neighbors); Generalized Linear Models (logistic regression, Poisson regression).

Topic 6: Linear model selection and regularization (subset selection, shrinkage methods: lasso and ridge regression, dimension reduction methods: principal components analysis (PCA), factor analysis).

Topic 7: Decision trees methods (bagging, random forests, boosting).

Topic 8: Smoothing techniques (splines, kernels, local weighted regression, loess).

Topic 9: Unsupervised learning (k-means clustering, principal components analysis).

Inclusive teaching statement:

- "I am committed to fostering an environment for learning that is inclusive for everyone regardless of gender identity, gender expression, sex, sexual orientation, race, ethnicity, ability, age, class, etc."
- "All students in the class, the instructor, and any guests should be treated with respect during all interactions."
- "It is my hope that our class will support diversity of experience, thought, and perspective."
- "Please feel free to contact me via email or in person to let me know about any experiences you have had related to this class that have made you feel uncomfortable."
- "I will continually strive to create inclusive learning environments and would therefore appreciate your support and feedback."
- "I welcome emails or in-person communications to let me know your preferred name or pronoun."

Land Acknowledgement

Here at Carleton University, it is important that we acknowledge that the land on which we gather is the traditional and unceded territory of the Algonquin nation.

Community Guidelines

The following values are fundamental to academic integrity and are adapted from the International Center for Academic Integrity*. In our course, we will seek to behave with these values in mind:

	As students, we will…	As a teaching team, we will…
Honesty	• Honestly demonstrate our knowledge and abilities on project and exams • Communicate openly without using deception, including citing appropriate sources	• Give you honest feedback on your demonstration of knowledge and abilities on project and exams • Communicate openly and honestly about the expectations and standards of the course through the syllabus, and with respect to project and exams
Responsibility	• Complete project on time and in full preparation for class • Show up to class on time, and be mentally/physically present • Participate fully and contribute to team learning and activities	• Give you timely feedback on your project and exams • Show up to class on time, and be mentally & physically present • Create relevant assessments and class activities
Respect	• Speak openly with one another, while respecting diverse viewpoints and perspectives • Provide sufficient space for others to voice their ideas	• Respect your perspectives even while we challenge you to think more deeply and critically • Help facilitate respectful exchange of ideas
Fairness	• Contribute fully and equally to collaborative work, so that we are not freeloading off of others • Not seek unfair advantage over fellow students in the course	• Create fair exams, and grade them in a fair, and timely manner • Treat all students equitably

Trust	• Not engage in personal affairs while on class time • Be open and transparent about what we are doing in class • Not distribute course materials to others without authorization	• Be available to all students when we say we will be • Follow through on our promises • Not modify the expectations or standards without communicating with everyone in the course
Courage	• Say or do something when we see actions that undermine any of the above values • Accept a lower or failing grade or other consequences of upholding and protecting the above values	• Say or do something when we see actions that undermine any of the above values • Accept the consequences (e.g., lower teaching evaluations) of upholding and protecting the above values

* This class statement of values is adapted from Tricia Bertram Gallant, Ph.D.

Statement on the Use of Generative AI

As our understanding of the uses of AI and its relationship to student work and academic integrity continue to evolve, students are required to discuss their use of AI in any circumstance not described here with the course instructor to ensure it supports the learning goals for the course.

Online Community Expectations for Social Platforms

With the growing use of social platforms (e.g., Discord) on campuses, it is important to keep in mind that university codes of conduct still apply to the behaviours of students online. Please be considerate and respectful while engaging with peers and remember that we are all humans, and that your words matter. If any student witnesses or experiences harassment, I encourage you to reach out to me. Alternatively, you can contact Ombuds Services or Carleton Equity and Inclusive Communities.

Online communities can be highly beneficial to students and can help to facilitate learning within the course. I encourage people to ask questions, learn from one another, and have open discussions about class material. That said, any acts of academic misconduct (i.e., cheating) will not be tolerated and will result in serious consequences ranging from a grade reduction to expulsion (*see academic integrity violations*).

- Examples of appropriate peer-to-peer sharing/learning vary from course to course. In this course appropriate peer-to-peer sharing includes: identifying the proper formula

to use, identifying an incorrect or missing step in a person's work, brainstorming potential reasons behind a concept, suggesting helpful sites and videos for learning a concept, posting your own work showing only a specific step or process for illustrative purposes (note: this is very different from posting your work and solution for others to simply copy).

- Examples of unacceptable peer-to-peer sharing: Posting or sharing the answers, indicating which answers are correct on research project, sharing links to solutions, posting your own complete work for a question/solution.

I ask that you read and follow the list of guidelines below (moderators may re-post, if helpful):

General Rules & Guidelines

- No inappropriate or sexually explicit profile pictures.
- No inappropriate or offensive nicknames.
- No inviting new people to join the server.
- Moderators reserve the right to delete or edit any posts.
- Moderators reserve the right to use their own discretion regardless of the written rules.
- No private/direct messaging moderators (please email the TAs or me with private matters).
- No NSFW or illegal content.
- No political discussions.
- No religious discussions.
- No sexual discussions.
- No bullying or harassment (towards other students, teaching assistants, or instructors).
- No encouraging harassment.
- No questioning actions of the mods (email if there is confusion or a problem you would like to discuss).
- No offensive or inappropriate language
- No racism, sexism, or hate speech of any kind.

There may be specific situations not covered by these rules, and there may also be certain cases where a rule does not apply. If you are concerned, confused, or conflicted over

something, please reach out to a TA or me through email for help. Let's do our best to support one another in this class and keep the online experience a safe, inclusive, and positive experience for everyone.

Assessment in this Course

Research about learning strongly suggests that the most important factor in learning is doing the work of reading, writing, recalling, practicing, synthesizing, and analyzing. Learning happens best when people actively engage material on a consistent basis, and that is why we have high standards in this course. We are confident that, with appropriate effort, you **all** can meet those standards.

Course Outline Evaluation Components for EP course:

There are 3 tests in this course and research project. See the marking rubric.

TEST	GRADE VALUE	TOPIC	TIME
MIDTERM TEST 1	20%	Covers lectures and readings up to February 4	February 11
MIDTERM TEST 2	20%	Covers lectures and readings up to March 18	March 25
ATTENDANCE AND PARTICIPATION	5%	To be eligible, students must attend the entire class and actively participate throughout	
RESEARCH PROJECT	15%	Choose your own topic	April 01
FINAL EXAM	40%	Comprehensive covers all lectures and readings	April 11

Please note that a passing grade for all EP courses is "B¯"

Missed tests

There is no make-up for missing the midterm test. Students are permitted to miss a maximum of one test due to extenuating circumstances as defined by the Academic Consideration Policy. These students must complete the Academic Consideration for Coursework Request Form within three business days of the original test. Students who correctly follow this procedure will have the missing test grade replaced by their grade on the final exam. Failure to follow this procedure will result in a grade of 0% on the missed test. Students who miss a second test will be required to meet with the instructor to determine whether they will be permitted to remain in the course. To defer a final exam, students must apply through the Registrar's Office.

Research Project

There will be a group project (**report and oral presentation**) in this course.

- You will work in groups of 2 persons to complete a data science project starting as early as you can.
- Each group need to formulate a research question of its choice, and then use a suitable dataset to answer this question.
- Below a list of some suggested sources to get data sets, but you can use another data that interests you.
    - Kaggle data science and machine learning dataset: https://www.kaggle.com/datasets
    - Water survey of Canada: https://www.canada.ca/en/environment-climate-change/services/water-overview/quantity/monitoring/survey.html
    - UCI Machine Learning Repository
    - Climate and Weather Data: https://climate.weather.gc.ca/
    - City of Ottawa data set. You may try other cites (Toronto, Vancouver, Montreal, etc.)
    - Stock market and finance: https://ca.finance.yahoo.com/, https://www.gold.org/goldhub/data/gold-prices, https://coinmarketcap.com/, https://www.investing.com/
    - Coronavirus Media Monitor: https://coronavirus.ravenpack.com/
- It is expected that the group members will collaborate on to produce a professional report and presentation that demonstrates the course concepts.
- The report should be submitted as an RMarkdown document (max 20 pages excluding the citations and code). Make sure that you include a tidy code that can be used to reproduce all the results you will get.
- Only one member of the team needs to submit electronically (cc the other students sharing this study) the final report. This report is two files:
    - the source R Markdown document (.rmd file)
    - the rendered output document (.html or .pdf file)
- Each report should include the following sections:
    - Title
    - Introduction
        - Provide literature review about the topic so that someone unfamiliar with it will be prepared to understand the rest of your project.
    - Methods and Output Results
        - Describe the methods you used to perform the analysis from beginning to end that you used to answer the research question.
        - Interpret/explain the results you obtain. For example, it's not enough to just say "We fitted a linear model with these covariates, and my adjusted R-square is 0.94".
        - Ensure the tables and/or figures are labeled with tables and/or figures numbers.
    - Discussion and Conclusion
        - Summarize what you found and discuss the implications/impact of your findings.

        - Discuss whether your results agree what you expected to find.
        - Discuss whether and how your final model could be improved?
        - Discuss future questions/research that your study could lead to.
    - References
        - Include all citations that you used in your project. The citation format is your choice – just be consistent. Make sure that you also include the source of the data set that you used in your study.

- **The due date to submit the report is April 19, 2026, by 11:59 pm.**
- Late submission will lose 50% per day. A project report can no longer be submitted one day following its initial deadline.
- The project accounts for 15% (**10% for the report and 5% for the presentation**) of the course grades:
- **For the report:**
    - I will evaluate the group's overall performance using the items 1-5 of the project report rubric (given in the next page).
    - The students will evaluate their group members based on their contributions to their research project using item 6 of this rubric. The average score of this item will be calculated.
    - Your final grade of the report project (out of 10 points) is calculated based on the sum scores that you will get for the items 1-6 of this rubric (score from the instructor + score form the other students in your group).
- **For the oral presentation:**
    - Each group will be given maximum 15 minutes to present, and all students in the group are expected to participate in the presentation.
    - The day for the oral presentation will be **April 1st during the class time**.
    - I will evaluate individually, each presenter, using the items 1-5 of the project presentation rubric (given in the next page).
    - The students form the other groups will evaluate each presenter using the same project presentation rubric that I will use. The average score of this item will be calculated.
    - Your final grade of the oral presentation of this project (out of 5 points) is calculated based on the weighted average score (0.7 * score from the instructor + 0.3 * score form the students in the entire class).

Project Rubric (for the report)

Criteria	Excellent (9-10)	Good (7-9)	Satisfactory (5-7)	Needs Improvement (3-5)	Unsatisfactory (0-3)	Score
Problem Understanding and Significance	Clearly understands and articulates the problem's importance, scope, and relevance.	Understands the problem and its significance but may lack some depth or clarity.	Shows basic understanding of the problem's significance but lacks clear articulation.	Demonstrates limited understanding of the problem's significance and relevance.	Fails to grasp the problem's significance.	
Analysis and Interpretation	Analyzes data with depth, applies advanced techniques, and derives meaningful insights.	Conducts sound data analysis and provides reasonable interpretations.	Performs basic data analysis but lacks in-depth insights.	Provides limited or incorrect analysis and interpretations.	Presents inadequate or incorrect analysis.	
Collaboration and Teamwork	Demonstrates excellent collaboration, active participation, and effective communication.	Collaborates well, contributes to discussions, and communicates effectively.	Participates but may occasionally lack active engagement or communication.	Shows limited engagement and struggles with communication.	Fails to collaborate and communicate within the team.	
Documentation and Report Quality	Provides thorough, well-organized documentation with clear explanations and references.	Offers organized documentation with clear explanations.	Provides basic documentation but lacks organization or clarity.	Offers minimal documentation that is unclear or disorganized.	Fails to provide proper documentation.	
Creativity and Innovation	Demonstrates exceptional creativity and introduces innovative elements to the project.	Shows creativity and incorporates some innovative aspects.	Displays basic creativity but lacks significant innovation.	Lacks creativity and innovation in the project approach.	Fails to demonstrate creativity or innovation.	
Peer Evaluation	Provides thoughtful, constructive, and accurate feedback on team members' contributions.	Offers constructive feedback on team members' contributions.	Provides feedback, but it may lack depth or accuracy.	Provides limited feedback with little insight or accuracy.	Offers no or irrelevant feedback.	
Total grade out of 10 points = (your total score points from the 6 criteria) ÷ 6						

Project Rubric (for the oral presentation)

Criteria	Very Good/Excellent (9-10)	Satisfactory/Good (6-9)	Needs Improvement (3-6)	Unsatisfactory (0-3)	Score
Organization & Structure	Presentation is well-organized. Ideas flow logically.	Presentation is adequately organized, but some sections could be better structured.	Presentation lacks clear organization. Ideas are disconnected.	Presentation is extremely disorganized and difficult to follow.	
Visual Aids	well-designed and add value to the presentation. They effectively support key points.	Visual aids are present but may not consistently enhance the presentation. They somewhat support key points.	Visual aids are present but do not add significant value to the presentation. They minimally support key points.	Visual aids are absent or confusing.	
Time Management	Presentation is within the time limit and follows a nice format precisely.	Presentation is slightly over/under the time limit and somewhat follows a format.	Presentation significantly exceeds or falls short of the time limit. Deviates noticeably from the format.	Presentation is excessively long/short and disregards the prescribed format.	
Q&A Handling	Handles audience questions well. Provides thorough, well-reasoned responses that enhance the presentation.	Responds to audience questions adequately, but some answers lack depth or accuracy.	Struggles to address audience questions. Answers are often unclear or incorrect.	Unable to respond to audience questions satisfactorily.	
Overall Impression	The presentation is very good and demonstrates a high level of competence.	The presentation is generally satisfactory, meeting the basic requirements.	The presentation is below average and lacks polish.	The presentation is extremely weak and fails to meet expectations.	
Total grade out of 5 points = (your total score points from the 5 criteria)					

You may check the following noteworthy links when you conduct your research:

- Library's Homepage - https://library.carleton.ca/
  - Search box in middle (Omni) for broad searches and specific book/article titles
- Library Guide for Mathematics and Statistics - https://library.carleton.ca/guides/subject/mathematics-and-statistics
- Suggest a book purchase – https://library.carleton.ca/forms/purchase-suggestion
- Putting material on Course Reserve - https://library.carleton.ca/services/reserves-faculty-and-instructors
- Citation Management - https://library.carleton.ca/guides/help/citation-management
- Copyright - https://library.carleton.ca/copyright-carleton
- Research Data Management - https://library.carleton.ca/services/research-data-management
- Scholarly Communications - https://library.carleton.ca/services/scholarly-communications
  - Including funding, open access publishing, research impact

For more information, you may contact:

Scott Turner
STEM Librarian, Research Support Services
238 MacOdrum Library
Carleton University, Ottawa
scott.turner@carleton.ca

Mental Health

If you are struggling, please do not hesitate to reach out. I am happy to listen, and/or direct you to resources that might help. In terms of class, if you need extra help or missed a lesson, don't stress! Email me and we will set a time to meet. I'll work with you, I promise. Remember that Carleton also offers an array of mental health and well-being resources, which can be found here.

University Policies

In accordance with the Carleton University Undergraduate Calendar Regulations, the letter grades assigned in this course will have the following percentage equivalents:

A+ = 90-100	B+ = 77-79	C+ = 67-69	D+ = 57-59
A = 85-89	B = 73-76	C = 63-66	D = 53-56
A- = 80-84	B- = 70-72	C- = 60-62	D- = 50-52

F = <50
WDN = Withdrawn from the course

DEF = Deferred

Academic Accommodations

Carleton is committed to providing academic accessibility for all individuals. You may need special arrangements to meet your academic obligations during the term. The accommodation request processes, including information about the Academic Consideration Policy for Students in Medical and Other Extenuating Circumstances, are outlined on the Academic Accommodations website (students.carleton.ca/course-outline).

Academic Accommodations for Students with Disabilities

If you have a documented disability requiring academic accommodations in this course, please contact the Paul Menton Centre for Students with Disabilities (PMC) at 613-520-6608 or pmc@carleton.ca for a formal evaluation or contact your PMC coordinator to send your instructor your Letter of Accommodation at the beginning of the term. You must also contact the PMC no later than two weeks before the first in-class scheduled test or exam requiring accommodation (if applicable). After requesting accommodation from PMC, meet with your instructor as soon as possible to ensure accommodation arrangements are made. For more details, visit the Paul Menton Centre website.

Addressing Human Rights Concerns

The University and all members of the University community share responsibility for ensuring that the University's educational, work and living environments are free from discrimination and harassment. Should you have concerns about harassment or discrimination relating to your age, ancestry, citizenship, colour, creed (religion), disability, ethnic origin, family status, gender expression, gender identity, marital status, place of origin, race, sex (including pregnancy), or sexual orientation, please contact the Department of Equity and Inclusive Communities at equity@carleton.ca.

Religious Obligations

Please contact me with requests for academic accommodation during the first two weeks of class, or as soon as possible after the need for accommodation is known to exist. For more details, please review the Student Guide to Academic Accommodation (PDF, 2.1 MB).

Survivors of Sexual Violence

As a community, Carleton University is committed to maintaining a positive learning, working and living environment where sexual violence will not be tolerated, and where survivors are supported through academic accommodations as per Carleton's Sexual Violence Policy. For

more information about the services available at the university and to obtain information about sexual violence and/or support, visit: https://carleton.ca/sexual-violence-support/

Accommodations for Missed Work

Carleton recognizes that students may experience unexpected, temporary incapacitation (i.e., illness, injury, or extraordinary circumstances outside of a student's control). As a result, Carleton has put into place a protocol for students to apply for accommodations using a self-declaration form in the event of missed work. The form can be found at: https://carleton.ca/registrar/wp-content/uploads/self-declaration.pdf. Note that these forms should be used for short-term concerns related to missed work; if you are experiencing chronic, ongoing challenges which necessitate a broader solution, I recommend reaching out to the Paul Menton Centre and/or the Care Support team.

For Pregnancy

Please contact me with any requests for academic accommodation during the first two weeks of class, or as soon as possible after the need for accommodation is known to exist. For more details, please review the Student Guide to Academic Accommodation (PDF, 2.1 MB).

Accommodation for Student Activities

Carleton University recognizes the substantial benefits, both to the individual student and for the university, that result from a student participating in activities beyond the classroom experience. Reasonable accommodation must be provided to students who compete or perform at the national or international level. Please contact me with any requests for academic accommodation during the first two weeks of class, or as soon as possible after the need for accommodation is known to exist. For more details, see the Senate Policy on Accommodation for Student Activities (PDF, 25KB).

Academic Integrity

Academic Integrity is upholding the values of honesty, trust, respect, fairness, responsibility, and courage that are fundamental to the educational experience. Carleton University provides supports such as academic integrity workshops to ensure, as far as possible, that all students understand the norms and standards of academic integrity that we expect you to uphold. Your teaching team has a responsibility to ensure that their application of the

Academic Integrity Policy upholds the university's collective commitments to fairness, equity, and integrity.
(Adapted from Carleton University's Academic Integrity Policy, 2021).

Examples of actions that do not adhere to Carleton's Academic Integrity Policy include:

- Plagiarism
- Accessing unauthorized sites for project or tests
- Unauthorized collaboration on project and exams
- Using artificial intelligence tools such as ChatGPT when your assessment instructions say that it is not permitted

Please review the checklist linked here to ensure you understand your responsibilities as a student with respect to academic integrity and this course.

Sanctions for Not Abiding by Carleton's Academic Integrity Policy

A student who has not upheld their responsibilities under Carleton's Academic Integrity Policy may be subject to one of several sanctions. A list of standard sanctions in science can be found here.

Additional details about this process can be found on the Faculty of Science Academic Integrity website. Students are expected to familiarize themselves with and follow the Carleton University Student Academic Integrity Policy. The Policy is strictly enforced and is binding on all students.

Student Rights & Responsibilities

Students are expected to act responsibly and engage respectfully with other students and members of the Carleton and the broader community. See the 7 Rights and Responsibilities Policy for details regarding the expectations of non-academic behaviour of students. Those who participate with another student in the commission of an infraction of this Policy will also be held liable for their actions.

Student Concerns

If a concern arises regarding this course, **your first point of contact is me**: Email or drop in during student hours and I will do my best to address your concern. If I am unable to address your concern, the next points of contact are (in this order):

Note: You can also bring your concerns to Ombuds services.

Assistance for Students

Academic and Career Development Services: http://carleton.ca/sacds/

Writing Services: http://www.carleton.ca/csas/writing-services/

Peer Assisted Study Sessions (PASS): https://carleton.ca/csas/group-support/pass/

Math Tutorial Centre: https://carleton.ca/math/math-tutorial-centre/

Science Student Success Centre: https://sssc.carleton.ca/

Important Dates for Winter 2026

Consult the Calendar website for a full list of important dates and the most updated information.